

2025 SISBID Unsupervised Lab

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Data Description

gdat is Gene Expression Data, $n = 445$ patients $\times$ $p = 353$ genes

- Only 353 genes with somatic mutations from COSMIC are retained

Data is Level III TCGA BRCA RNA-Sequencing gene expression data that have already been pre-processed according to the following steps:

- Reads normalized by RPKM
- Corrected for overdispersion by a log-transformation ($1 + \text{data}$)
- Short gene name labels are given as the column names

cdat is Clinical Data, $n = 445$ patients $\times$ $q = 6$ clinical features

- Subtype - denotes 5 PAM50 subtypes including Basal-like, Luminal A, Luminal B, HER2-enriched, and Normal-like
- ER-Status - estrogen-receptor status
- PR-Status - progesterone-receptor status
- HER2-Status - human epidermal growth factor receptor 2 status
- Node - number of lymph nodes involved
- Metastasis - indicator for whether the cancer has metastasized

Problems

Problem 1 - Dimension reduction

1a - Apply PCA, NMF, ICA and MDS, UMAP, and tSNE to this dataset. Compare and contrast the results using these methods.

1b - Relate the dimension reduction results with the clinical data. Is any clinical information reflected in the lower dimensional spaces?

1c - Overall, which dimension reduction method do you recommend for this data set and why?

Problem 2 - Clustering

2a - Apply various clustering algorithms such as K-means (explore different K), hierarchical clustering (explore different linkages), NMF, and biclustering. Compare the clustering results using these methods.

2b - Relate the clustering results with the clinical data. Can the clustering algorithm recover some of the clinical information such as cancer subtypes?

2c - (Optional) Validate your cluster findings.

2c - Overall, which clustering method(s) do you recommend for this data set and why?

Problem 3 - Multiple comparisons

3a - Identify important genes to differentiate ER positive and negative, PR positive and negative, HER2 positive and negative, metastasis status.

3b - Try different procedures to adjust for multiple comparisons.

3c - Examine the lists of genes identified using different methods for each clinical response. Which method is best? Why?

Problem 5 - Graphical models

5a - Use graphical models to explore interactions among genes. Are any of the well-connected genes related to patterns previously identified?

Problem 6 - Visualization

6a - Visualize this data using multiple approaches.

6b - Prepare the “best” visual summary of this data.

Problem 7 - Exploratory Data Analysis Summary

7a - What is the most interesting finding?

7b - Is this finding consistent and stable?

7c - Prepare a visual summary that best illustrates this interesting finding.

R scripts to help out with the BRCA case study Lab

Don't peek at this if you want to practice coding on your own!!

Load Data

```
load("UnsupL_SISBID_2025.Rdata")
library(ggplot2)
library(kknn)
library(GGally)
library(umap)
library(Rtsne)
library(igraph)
```

```
##
```

```
## Attaching package: 'igraph'
```

```
## The following objects are masked from 'package:stats':
##
##   decompose, spectrum
```

```
## The following object is masked from 'package:base':
##
##   union
```

```
library(huge)
```

Explore Data

```
dim(gdat)
```

```
## [1] 445 353
```

```
dim(cdat)
```

```
## [1] 445 6
```

```
# clinical data
table(cdat$Subtype)
```

```
##
##   Basal-like HER2-enriched   Luminal A   Luminal B   Normal-like
##           79           53           200           106           7
```

```
table(cdat$ER)
```

```
##
##           Indeterminate           Negative
##           2           100
##   Not Performed Performed but Not Available
##           2           2
##           Positive
##           339
```

```
table(cdat$PR)
```

```
##
##           Indeterminate           Negative
##           3           147
##   Not Performed Performed but Not Available
##           2           2
##           Positive
##           291
```

```
table(cdat$HER2)
```

```
##
##      Equivocal      Negative Not Available      Positive
##           5           370           4           65
```

```
table(cdat$Node)
```

```
##
##    0    1    2    3
## 221 146  54  23
```

```
table(cdat$Metastasis)
```

```
##
##    0    1
## 427  11
```

```
table(cdat$ER,cdat$PR)
```

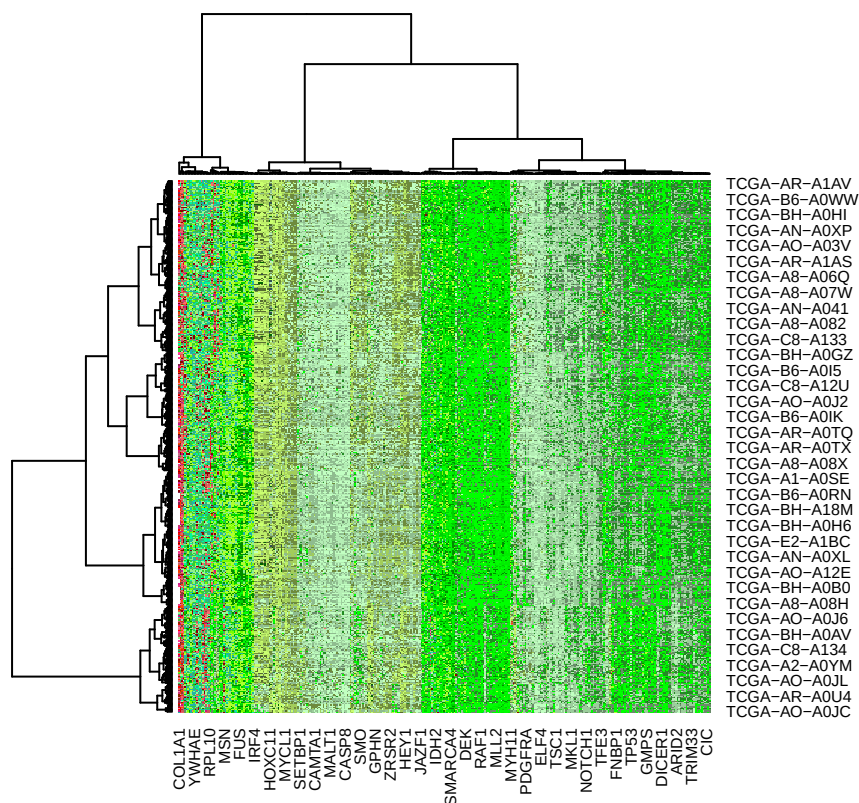
```
##
##              Indeterminate Negative Not Performed
## Indeterminate              0          1          0
## Negative              1          93          0
## Not Performed              0          0          2
## Performed but Not Available  0          0          0
## Positive              2          53          0
##
##              Performed but Not Available Positive
## Indeterminate              0          1
## Negative              0          6
## Not Performed              0          0
## Performed but Not Available  2          0
## Positive              0         284
```

Cluster Heatmap - biclustering

```
#cluster heatmap - biclustering
aa = grep("grey",colors())
bb = grep("green",colors())
cc = grep("red",colors())
gcol2 = colors()[c(aa[1:2],bb[1:25],cc[1:50])]
```

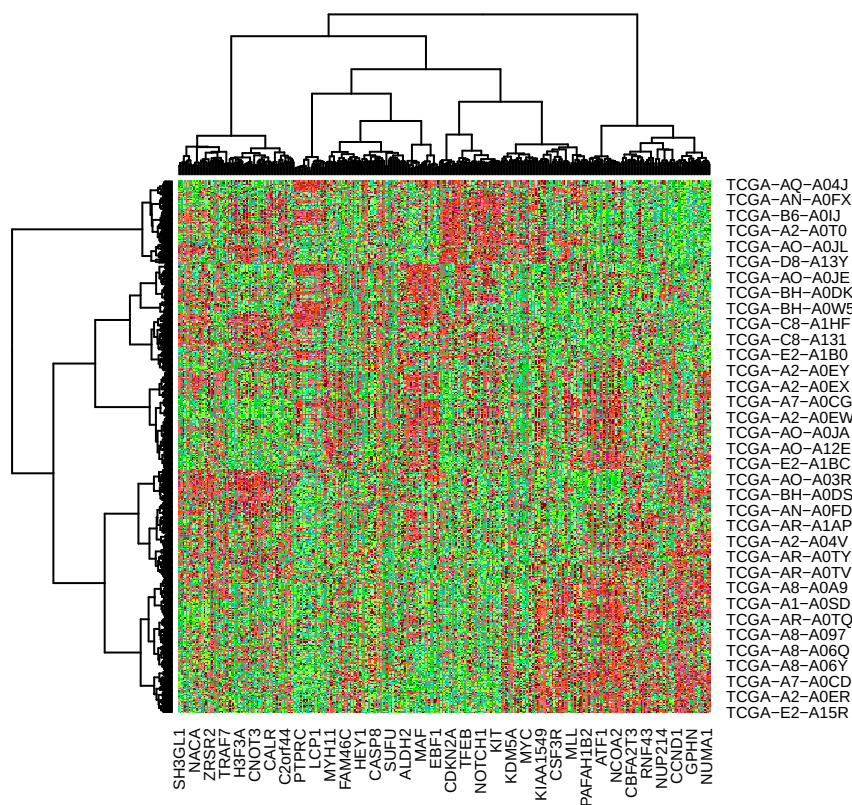
Without scaling

```
heatmap(gdat,col=gcol2,hclustfun=function(x)hclust(x,method="ward.D"))
```



With scaling

```
heatmap(scale(gdat),col=gcol2,hclustfun=function(x)hclust(x,method="ward.D"))
```

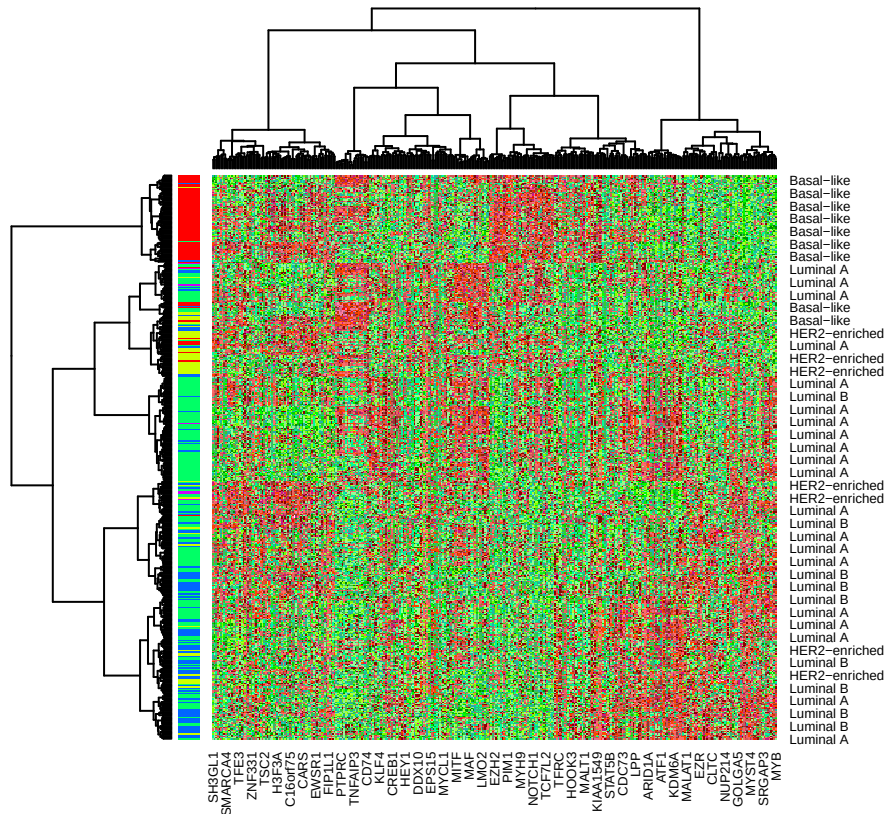


```

Cols=function(vec){cols=rainbow(length(unique(vec)))
  return(cols[as.numeric(as.factor(vec))])}

heatmap(scale(gdat),col=gcol2,hclustfun=function(x)hclust(x,method="ward.D"),labRow=cdat$Subtype,RowSide

```



Dimension Reduction

PCA

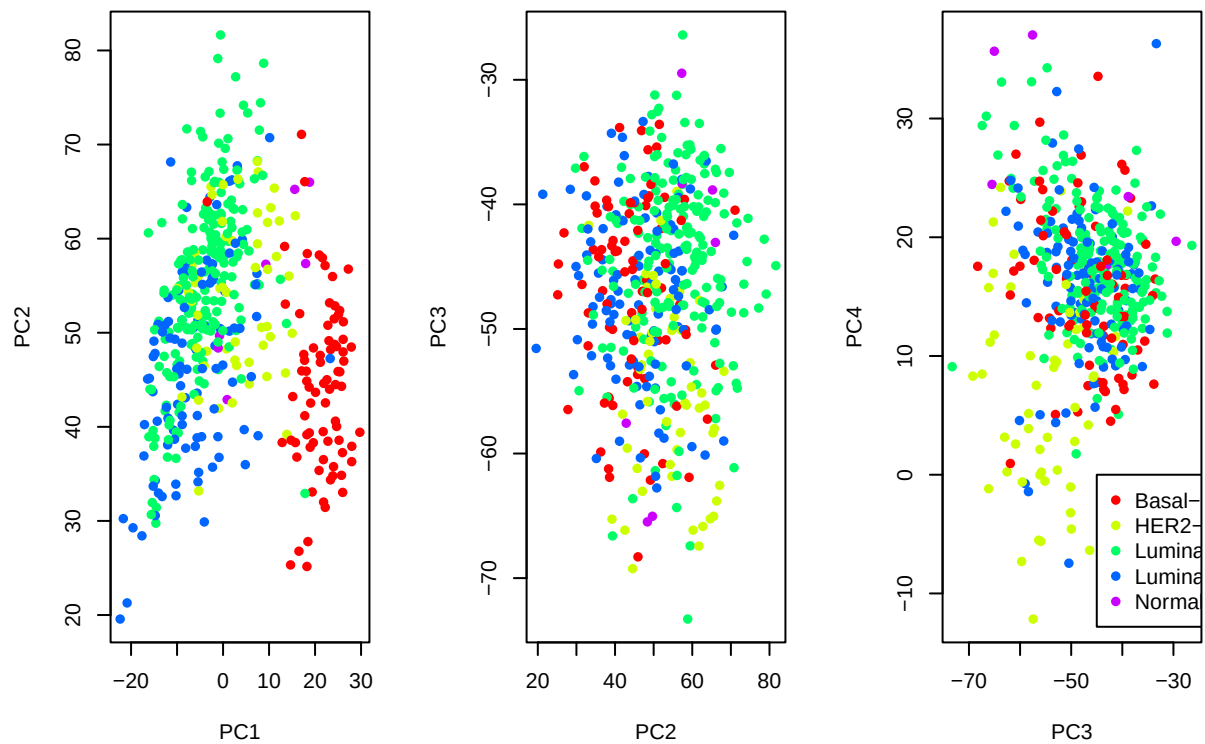
```

Cols=function(vec){cols=rainbow(length(unique(vec)))
  return(cols[as.numeric(as.factor(vec))])}

sv = svd(scale(gdat,center=TRUE,scale=FALSE))
V = sv$v
Z = gdat%*%V

K = 3
pclabs = c("PC1", "PC2", "PC3", "PC4")
par(mfrow=c(1,K))
for(i in 1:K){
  j = i+1
  plot(Z[,i],Z[,j],pch=16,xlab=pclabs[i],ylab=pclabs[j],col=Cols(cdat$Subtype))
}
legend(-45,0,pch=16,col=rainbow(5),levels(cdat$Subtype))

```

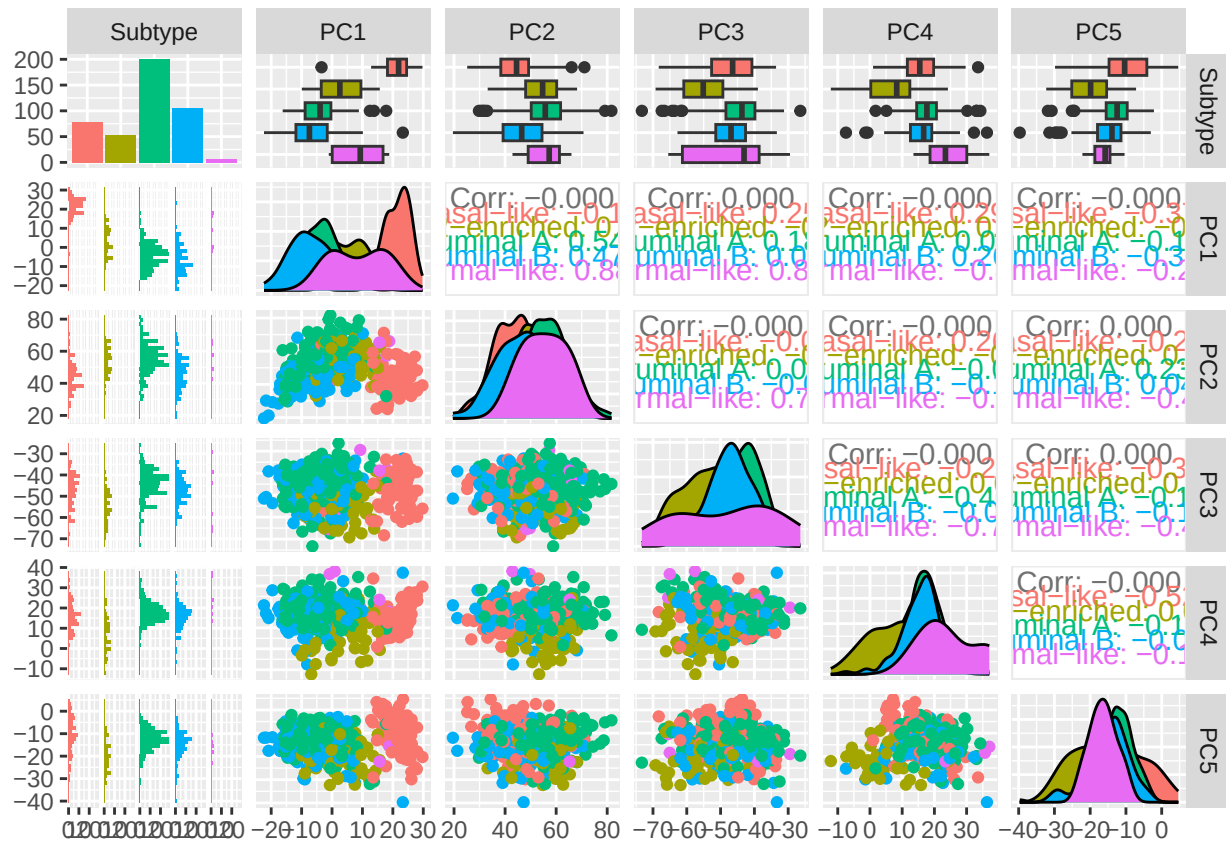


Pairs Plot

```
PC1<-as.matrix(Z[,1])
PC2<-as.matrix(Z[,2])
PC3<-as.matrix(Z[,3])
PC4<-as.matrix(Z[,4])
PC5<-as.matrix(Z[,5])
```

```
pc.df.cdat<-data.frame(Subtype = cdat$Subtype, PC1, PC2, PC3, PC4, PC5)
ggpairs(pc.df.cdat, mapping = aes(color = Subtype))
```

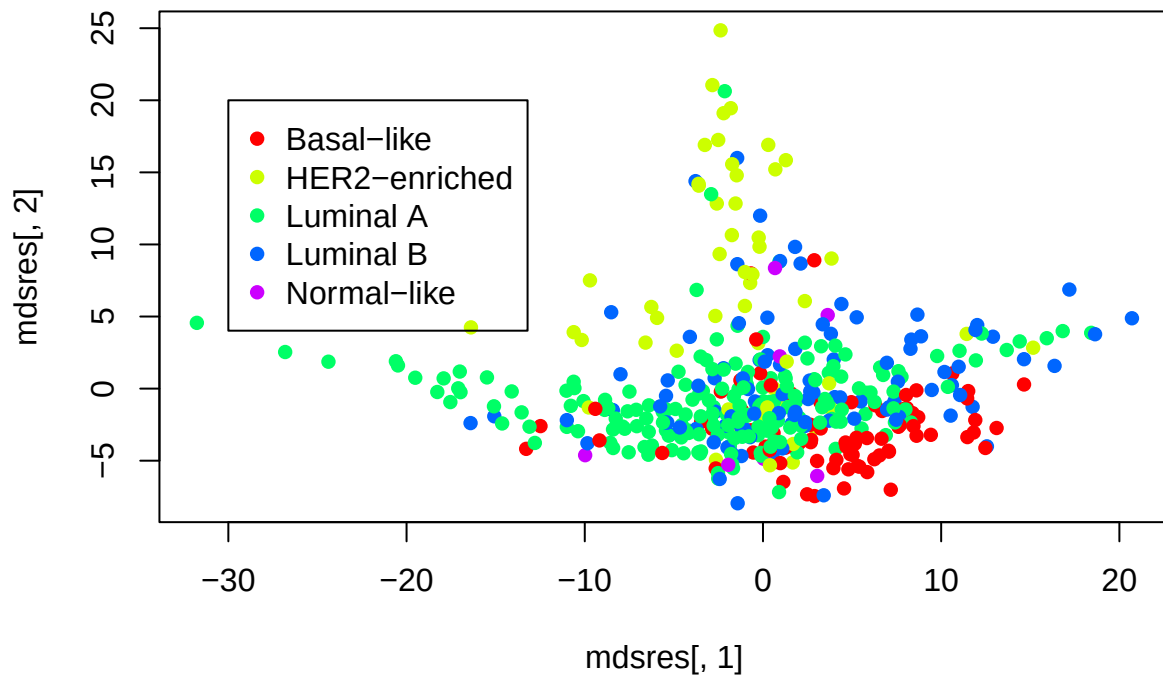
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



MDS

```
Dmat = dist(gdat,method="maximum")
mdsres = cmdscale(Dmat,k=2)
plot(mdsres[,1],mdsres[,2],pch=16,col=Cols(cdat$Subtype), main = "Dimension Reduction MDS")
legend(-30,20,pch=16,col=rainbow(5),levels(cdat$Subtype))
```


Dimension Reduction MDS



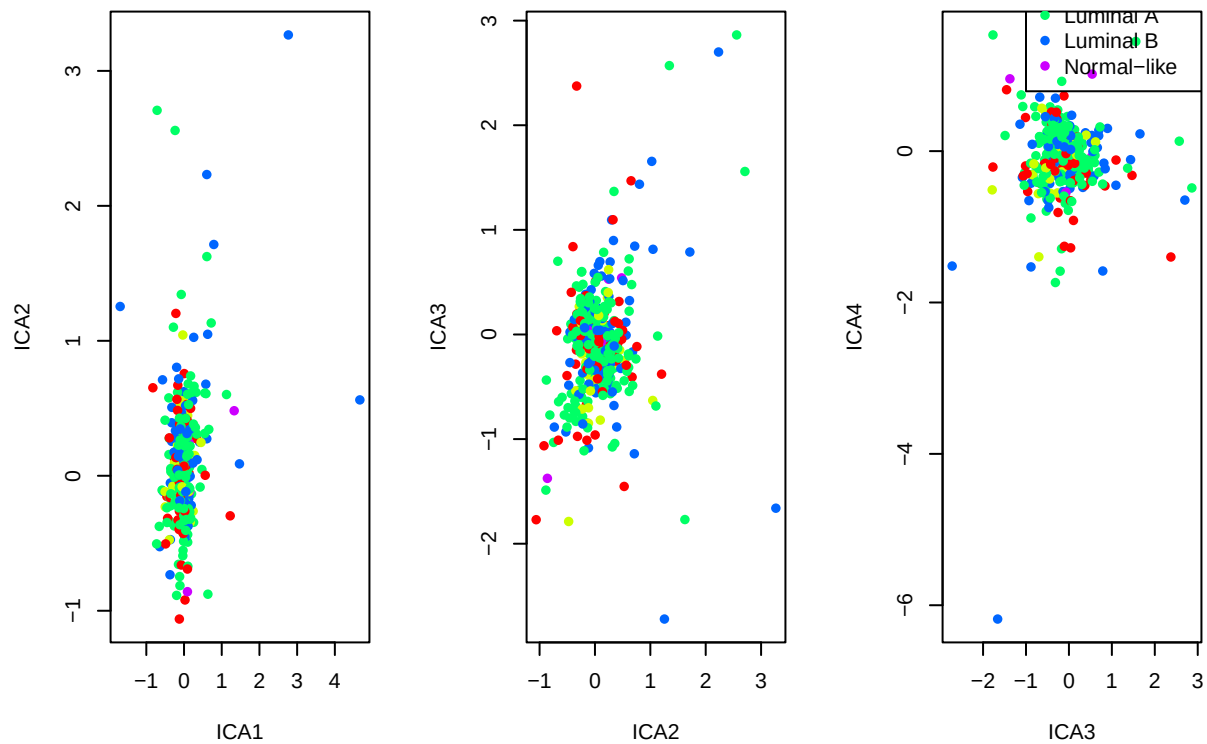
ICA

```
require("fastICA")
```

```
## Loading required package: fastICA
```

```
K = 4
icafit = fastICA(gdat,n.comp=K)

kk = 3
pclabs = c("ICA1","ICA2","ICA3","ICA4")
par(mfrow=c(1,kk))
for(i in 1:kk){
  j = i+1
  plot(icafit$A[i,],icafit$A[j,],pch=16,xlab=pclabs[i],ylab=pclabs[j],col=Cols(cdat$Subtype))
}
legend(-1,2.8,pch=16,col=rainbow(5),levels(cdat$Subtype))
```

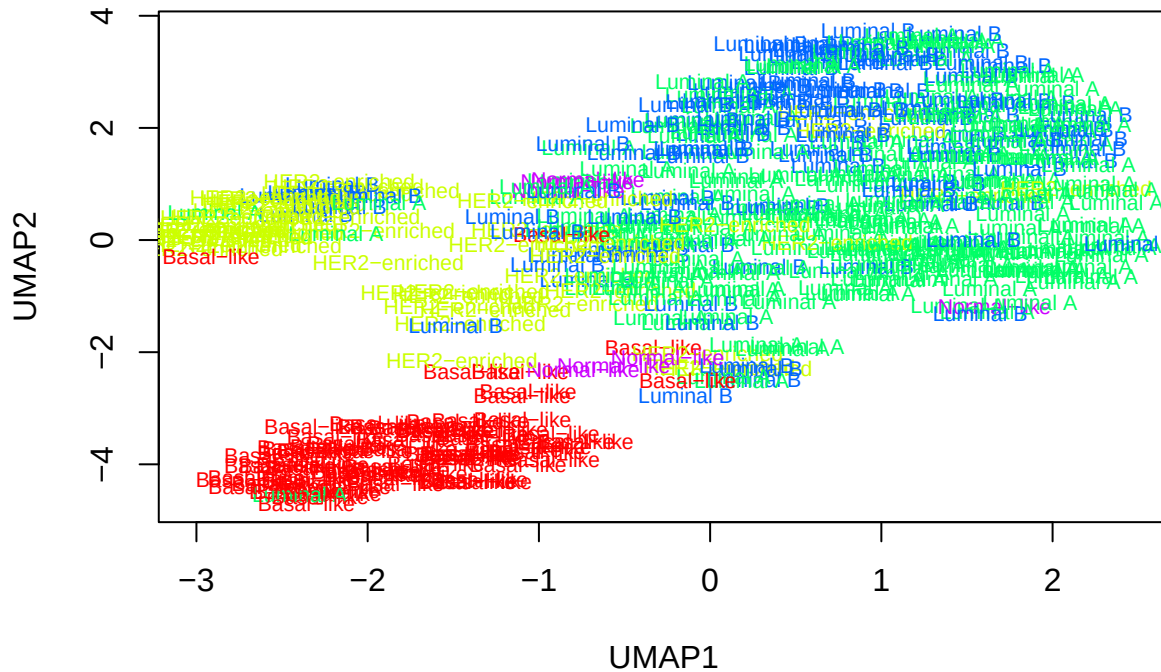


UMAP

```
gdat.umap = umap(gdat)
plot(gdat.umap$layout[,1], y = gdat.umap$layout[,2], type = "n", main = "UMAP", xlab = "UMAP1", ylab = "UMAP2")
text(gdat.umap$layout[,1], y = gdat.umap$layout[,2], type = "n", cdat$Subtype, col = Cols(cdat$Subtype), c
```

```
## Warning in text.default(gdat.umap$layout[, 1], y = gdat.umap$layout[, 2], :
## graphical parameter "type" is obsolete
```

UMAP



Clustering

K-means

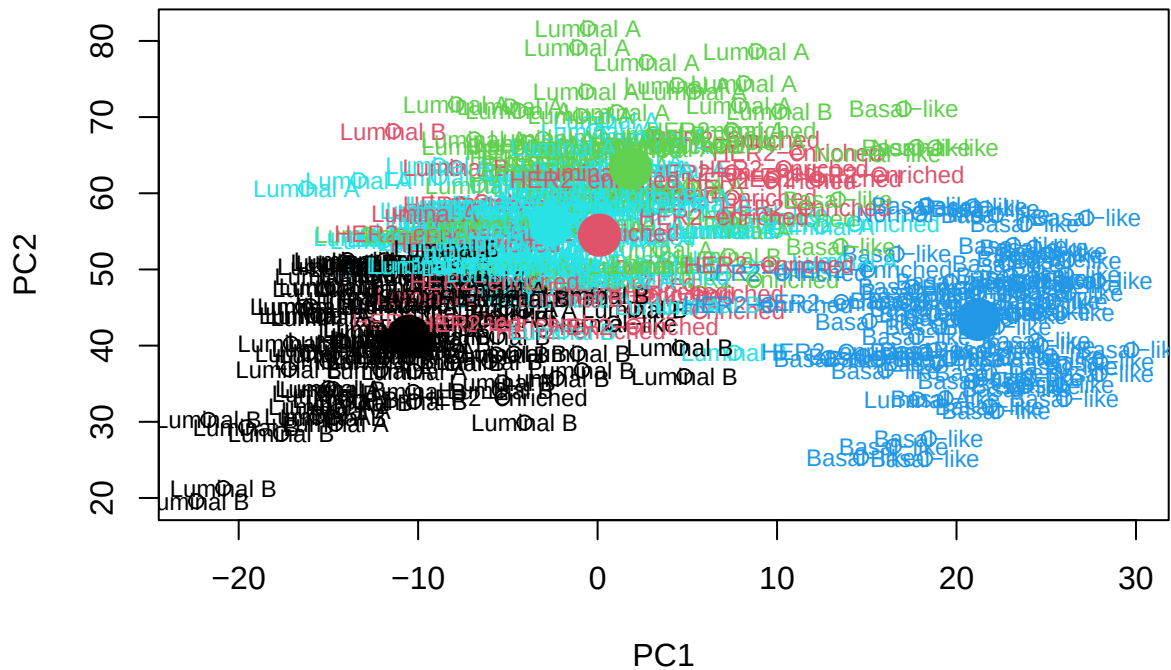
```
K = 5
km = kmeans(gdat,centers=K,nstart=25)
table(km$cluster,cdat$Subtype)
```

```
##
##      Basal-like HER2-enriched Luminal A Luminal B Normal-like
##  1           0              2         42         60           1
##  2           0             31          3          7           0
##  3           5              8         45         12           4
##  4          74              5          1          1           1
##  5           0              7        109         26           1
```

Plot Kmeans with labels

```
plot(Z[,1],Z[,2],col=km$cluster, main = "Plot Kmeans Clusters ", xlab = "PC1", ylab = "PC2")
text(Z[,1],Z[,2],cdat$Subtype,cex=.75,col=km$cluster)
cens = km$centers
points(cens%*%V[,1],cens%*%V[,2],col=1:K,pch=16,cex=3)
```

Plot Kmeans Clusters



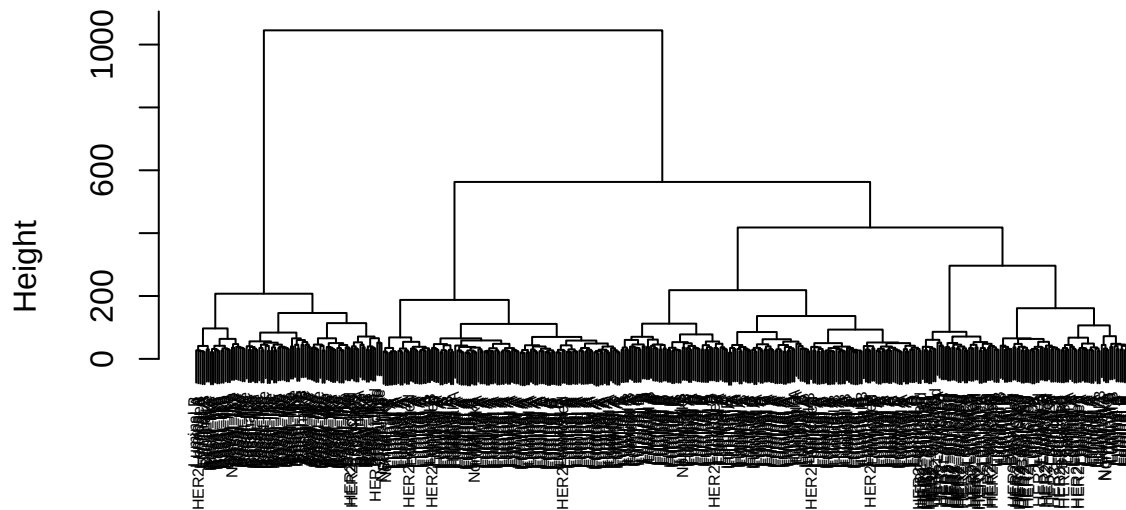
Hierarchical

```
#which linkage is the best?
#which distance metric is the best?

Dmat = dist(gdat)
com.hc = hclust(Dmat,method="ward.D")

plot(com.hc,labels=cdat$Subtype,cex=.5)
```

Cluster Dendrogram



Dmat
hclust (*, "ward.D")

```
res.com = cutree(com.hc,5)
table(res.com,cdat$Subtype)
```

```
##
## res.com Basal-like HER2-enriched Luminal A Luminal B Normal-like
##      1      1      3      95      11      3
##      2      0      4      73      65      1
##      3     75      4       5       4      1
##      4      0     27       3       7      0
##      5      3     15      24      19      2
```

Consensus Clustering with Hierarchical

```
#Note that ConsensusClusterPlus not available for R version 4.0.2
#results = ConsensusClusterPlus(gdat,maxK=6, reps=500, pItem=0.8, pFeature=1,
#clusterAlg="hc", distance="pearson", plot="png")
```

Look at results for first 5 clusters

```
#heatmap(results[[2]][["consensusMatrix"]][1:5,1:5])
```

Spectral Clustering

```
K = 5
s_gdat = specClust(gdat, centers=K, nn = 7, method = "symmetric", gmax=NULL)
```

```
plot(Z[,1],Z[,2],col=s_gdat$cluster, main = "Visualize Spectral Clusters", xlab = "PC1", ylab = "PC2")
text(Z[,1],Z[,2],cdat$Subtype,cex=.75,col=s_gdat$cluster)
```



```
x = gdat[cdat$ER=="Positive" | cdat$ER=="Negative",]
y.er = cdat$ER[cdat$ER=="Positive" | cdat$ER=="Negative"]
y.label = rep(1, length(y.er))
y.label[y.er == "Positive"]=2

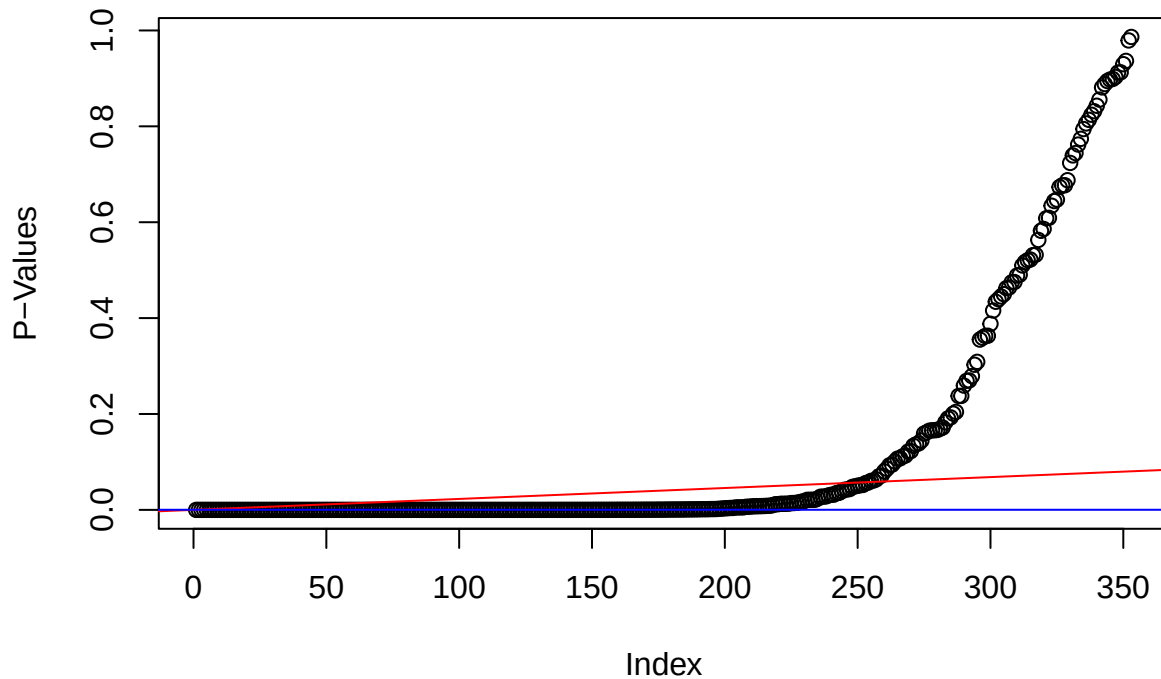
ps = NULL
for(i in 1:ncol(gdat)) ps = c(ps,
  t.test(x[y.label==1,i],x[y.label==2,i])$p.value)
fdrs.bh = p.adjust(ps, method="BH")

cat("Number of Tests significant with alpha=0.1 using Bonferroni correction:",
sum(ps<0.1/length(y.label)), fill=TRUE)
```

```
cat("Number of Tests with FDR below 0.1:",  
sum(fdrs.bh<0.1), fill=TRUE)
```

14

```
plot(sort(ps,decreasing=FALSE),ylab="P-Values")
#BH procedure
abline(a=0, b=0.1/length(y.label),col="red")
#Bonferroni
abline(a=0.1/length(y.label), b=0,col="blue")
```



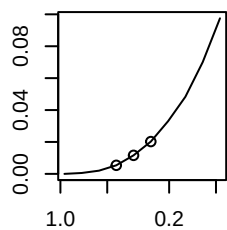
Graphical models - how are genes related?

```
# use huge package
neth = huge(gdat,method="mb")
```

```
## Conducting Meinshausen & Buhlmann graph estimation (mb)....done
```

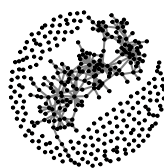
```
plot(neth)
```

Sparsity vs. Regularization

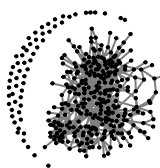


Regularization Parameter

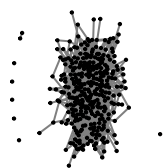
lambda = 0.437



lambda = 0.339



lambda = 0.262



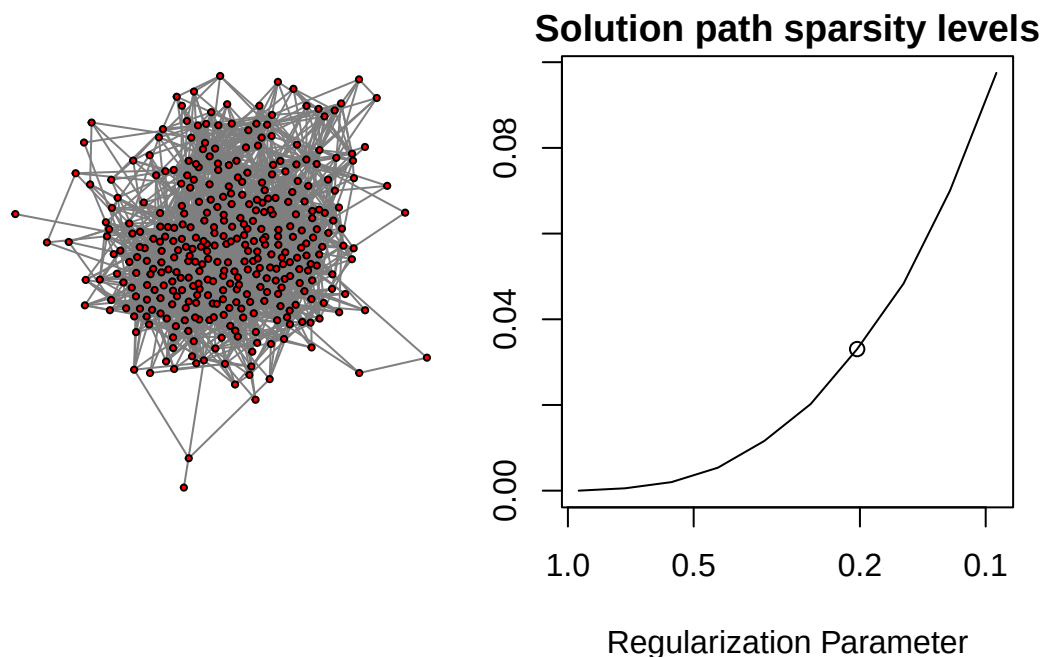
```
## stability selection with huge
net.s <- huge.select(neth, criterion="stars")
```

```
## Conducting Subsampling...in progress:5% Conducting Subsampling...in progress:10% Conducting Subsam
```

```
net.s
```

```
## Model: Meinshausen & Buhlmann Graph Estimation (mb)
## selection criterion: stars
## Graph dimension: 353
## sparsity level 0.03304468
```

```
plot(net.s)
```



```
#larger lambda
mat <- neth$path[[2]]
neti <- as.undirected(graph_from_adjacency_matrix(mat))
```

```
## Warning: 'as.undirected()' was deprecated in igraph 2.1.0.
## i Please use 'as_undirected()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
plot(neti, vertex.label=colnames(gdat), vertex.size=2, vertex.label.cex=1.2, vertex.label.dist=1, layout=lay
```